MedAgent Clinical Decision Support

Case ID: case_dd092630ce | **Status:** APPROVE | **Physician:** Dr. Alex Vance | **Date:** 2026-10-13 08:41 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: 68, Sex: male
Notes: 68yo male with 4-day history of productive cough with purulent rust-colored sputum, right-sided pleuritic chest pain, and high-grade fever (38.8°C). Dyspnea on mild exertion. Crackles audible at right lower lung base. No prior immunosuppression.

Clinical Summary / Impression

Patient evaluation demonstrates clinical features consistent with Community-Acquired Pneumonia. Diagnostic synthesis is supported by 5 key multimodal findings. Differential considerations include Sepsis / Severe Systemic Infection, Acute Kidney Injury. Diagnostic workup and clinical monitoring are indicated to confirm etiology and guide definitive management.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	Community-Acquired Pneumonia	High	fever, cough, sputum, crackles, pleuritic	S2, S8
2	Sepsis / Severe Systemic Infection	Moderate	fever, Leukocytosis with Neutrophilia, Raised Inflammatory Markers, Renal Impairment	S1, S3
3	Acute Kidney Injury	Moderate	Renal Impairment	S7
4	Pleural Effusion	Moderate	pleuritic, dyspnea, Raised Inflammatory Markers	S_Pleura_1, S_Pleu
5	Chronic Kidney Disease	Moderate	Renal Impairment	S_Chroni_1, S_Chro

Laboratory Findings & Identified Patterns

Identified Clinical Patterns: Leukocytosis with Neutrophilia (soon); **Raised Inflammatory Markers** (routine); **Renal Impairment** (soon)

Physician Attestation: I, Dr. Dr. Alex Vance, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **APPROVE**.
Reviewer Notes: No specific notes recorded.
Signature: _____ **Date:** 2026-10-13